Random Lattice Graphs

October 20, 2026

Abstract

We define a finite and bounded 2-dimensional lattice graph of radius r, explore general graphs constructed over the bounded set $V_r = \{v \in \mathbb{Z}^2 : \|v\|_\infty \leq r\}$, and explore randomized graph construction over this space.

1 The Bounded Lattice Graph

The 2-dimensional integer lattice is the set $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$. From this set, we can build the infinite lattice graph $\mathbb{L}^2$ with vertices $V = \mathbb{Z}^2$ and edges connecting any two vertices u, v so long as $d(u, v) = 1$ with respect to the standard metric. Now, we define the finite, bounded subgraph $\mathbb{L}^2(r) \subset \mathbb{L}^2$:

$$\mathbb{L}^2(r) = \mathbb{L}^2\left[\{v \in \mathbb{Z}^2 : \|v\|_\infty \leq r\}\right], r \in \mathbb{Z}^+ \tag{1}$$

We may choose to call this the lattice graph of radius r, and we may think of its vertex set V_r as the collection of points $(x, y) \in \mathbb{Z}^2$ for which $x, y \in [-r, r]$. The geometry of the vertex set alone is that of a discrete box bounded by the $\mathbf{L}^\infty$ norm. For the entire graph, however, we use the standard notion of distance and edge length/weight implied on $\mathbb{L}^2$. Particularly, we let the $\mathbf{L}^1$ metric (Manhattan distance) define the distance between any two vertices.

From this initial definition, we can immediately see that $\mathbb{L}^2(r)$ is finite and bounded by $\mathbf{L}^1$ metric. As for the vertex set, it is clearly finite with $|V_r| = (2r + 1)^2 < \infty$. We know that any node $u \in \mathbb{L}^2$ has degree 4. In $\mathbb{L}^2(r)$, a node's degree is always less than or equal to 4, depending on if it lies on the edge or center of the graph, thus the edge set E_r must be finite as well:

$$|E_r| < \max_{u \in V_r} \deg(u) \cdot |V_r| \leq 4|V_r| < \infty$$

As for the boundedness, the maximum distance between any two vertices in $\mathbb{L}^2(r)$ is achieved when they lie on opposing corners of the graph. In this case, for any $u, v \in \mathbb{L}^2(r)$,

$$d(u, v) \leq d_{\mathbf{L}^1}((-r, -r), (r, r)) = 4r < \infty$$

2 General Graphs over V_r

From $\mathbb{L}^2(r)$, we can use the vertex domain and the standard $\mathbf{L}^1$ distance to construct general bounded graphs. We can denote the graph generated over V_r by $G_r = (V, E, d_{\mathbf{L}^1})$, where $V \subseteq V_r$ and for all edges $e = (u, v) \in E$, $l(e) = d_{\mathrm{L}^1}(u, v)$. We can also define an unweighted version, where the length of edges and their exact path can be neglected, left only to fill the abstract notion of connectedness. For the standard, weighted case, we will assume that the $\mathbf{L}^1$ metric is standard for distances and edge lengths, so we will simply write $G_r = (V, E)$.

As the space V_r is bounded, it follows that any graph embedded in V_r must also be bounded. Similarly, $|V_r|$ is finite, so $V \subseteq V_r$ is too.

2.1 Isomorphisms and Embeddings

Let G be a graph that is not necessarily embedded in V_r. There are particular conditions that enable G to be represented as a graph embedded in some V_r. Assume $G = (V, E)$ is some unweighted graph, not embedded in V_r for some r. In this case, edge length is negligible, so as long as we choose a set of vertices $V' \subseteq V_r$ with $|V'| \geq |V|$, then we can build a graph G_r that is isomorphic to G. If the above condition is satisfied, there must exist an injective mapping $\phi : V \to V'$. As for the edges, if $(u, v) \in E_G$, we simply reconnect the images of these vertices in G_r to achieve the isomorphism:

$$(u, v) \in E_G \implies (\phi(u), \phi(v)) \in E_{G_r} \tag{2}$$

From the previous section, we know that $|V_r| = (2r + 1)^2$. Thus, for any graph G with n vertices that is not necessarily embedded in any V_r, we require $r \geq (\lceil(\sqrt{n} - 1)/2\rceil)$ to embed it into some finite, bounded lattice space.

Theorem. For all unweighted $G = (V, E)$ with $|V| = n$, there exists a graph G_r with $r \geq \lceil(\sqrt{n} - 1)/2\rceil$ such that $G \cong G_r$.

This process only fails if we wish to embed an infinite graph $G = (V, E)$ with $|V| = \infty$. In this case, we could only embed onto $V_\infty = \mathbb{Z}^2$.

Now, what if we want to embed a *weighted* graph into some lattice subspace V_r? From the above, we know how minimal r can be, but if we only considered the size of the vertex sets, then the condition $\max_{e \in E_G} l(e) > 4r$ would make an embedding into what we thought we required for r impossible. Thus, there are additional constraints we need to cover when dealing with weighted graphs in order to guarantee the existence of an isomorphic counterpart G_r.

Consider a weighted graph $G = (V, E, w)$ with $w : E \to \mathbb{Z}^+$ as its weight function. For reference, a graph embedded in V_r has weight function $w(e) = \|e\|_{\mathbf{L}^1}$. For starters, let us assume that G has only 2 nodes. Then, we require $r \geq \lceil(\sqrt{2} - 1)/2\rceil = 1$ to ensure we can at least map all of the nodes from G. With $r = 1$, we know the diameter of V_r, and equivalently, the maximum length of an edge from G_r, is $4r = 4$. However, if the edge in G connecting its only two nodes has weight $w(e) > 4$, we cannot embed G in V_1. So, we not only need $|V_r| \geq |V|$, but we also require:

$$\max_{e \in E} w(e) \leq 4r \implies r \geq \frac{1}{4} \max_{e \in E} w(e)$$

Thus, when finding the condition on r that would ensure the existence of an isomorphism $G \cong G_r$ for a weighted graph G, we should first choose r as described above, then increase it as needed to ensure we can map vertices.

3 Random Graphs Embedded in V_r

We begin the section by building a notion of randomized graph embedded in V_r. We must choose a random process that constructs both V and E. Let us construct $G_r = (V, E)$ using the following: $u \in V_r$ is apart of V with probability p_v, and an edge connects any two points/vertices $u, v \in V$ with probability p_e. From this simple random construction, the expected number of vertices n – equivalently $|V|$ – is:

$$\mathbb{E}[n] = p_v |V_r|$$

As for edges, let us first look at the expected degree of any particular vertex u. For any vertex u, there are $p_v|V_r| - 1$ choices for connection by the above. As an edge exists between any two vertices with probability p_e, we expect the degree of a given vertex to be:

$$\mathbb{E}[\deg(u)] = p_e(p_v|V_r| - 1)$$

Similar to the way we derive the expected number of vertices, we can consider the set of all possible edges, and multiply it by the probability p_e to achieve the expected number of edges for a random graph embedded in V_r generated by the random process described initially. For any graph with n vertices, the number of possible edges without duplication or self loops is $n(n-1)/2$. Let n be our expected number of vertices as previously described, then the expected number of edges must be:

$$\begin{aligned}\mathbb{E}\left[|E|\right] &= p_e\left(\frac{n(n-1)}{2}\right) \\ &= p_e\left(\frac{(p_v|V_r|)(p_v|V_r|-1)}{2}\right) \\ &= p_e\left(\frac{(p_v|V_r|)^2 - p_v|V_r|}{2}\right)\end{aligned}$$

We may also simply write it in terms of expected values:

$$\mathbb{E}[|E|] = p_e\left(\frac{\mathbb{E}[|V|]^2 - \mathbb{E}[|V|]}{2}\right)$$

Some immediate results to note are:

1. As $p_v \to 1$, $|V| \to |V_r|$
2. Fix $p_v > 0$. As $p_e \to 1, |E| \to \frac{1}{2}[(p_v|V_r|)^2 - p_v|V_r|]$